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COMPUTATIONAL FLUID DYNAMICS – A.A. 2024-2025

Impact of Ice Formation on the Aerodynamic Performance of the NACA 23012 Airfoil

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1. Abstract

Airfoils are highly optimized geometries, engineered to minimize drag and maximize lift. However, under specific atmospheric conditions, those shapes are altered by ice accretion, which ends up affecting their aerodynamic performance. This report investigates the impact of ice formation on the aerodynamic characteristics of the NACA 23012 airfoil. The study compares the clean configuration with three iced geometries, objects of test cases in the first AIAA Ice Prediction Workshop [1] and depicted here in Figure 1. Steady CFD simulations have been carried out with different turbulent-viscosity and transition models, on the basis that RANS methods do offer a satisfactory enough description of high-Reynolds-number flows, even when compared with more sophisticated approaches (see [8]); the results obtained are compared with each other showing the actual degradation of aerodynamic performance. Special attention is paid to Case-241, for which the impact of the actual roughness of ice has been studied. The simulations were performed using the SU2 solver, with grid generation carried out using Gmsh and Fidelity Pointwise. In an attempt to overcome some problems that arose during this research, the feasibility of generating grids for complex geometries using a combination of Gmsh and MATLAB® was investigated.

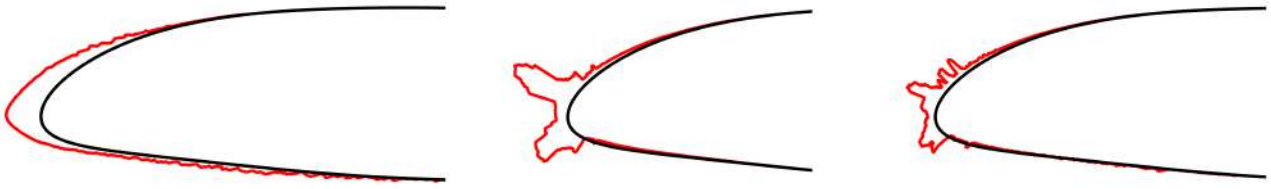


Figure 1: Comparison between the different iced geometries: Case-241 on the left, Case-242 in the center, Case-251 on the right.

2. Numerical simulation methodology

2.1. Mesh construction

The computational domain has been set as a circular-shaped domain and the chosen mesh was an hybrid one, despite computational costs. In order to capture the information inside the boundary layer (BL), structured mesh has been used, while the unstructured triangular mesh was preferred everywhere else in the domain. The meshing process has been divided into three phases: the first phase focuses

on the far-field characterization, the second phase deals with the elements distribution on the profile while the last one is the study of the grid convergence.

2.1.1 Far field independence

The first phase involves two subphases: determining the distance of the *inner far field* by fixing the appropriate spacing, as well as the spacing and distance for the *outer far field*. This entire step was accomplished in relation to clean airfoil simulations.
The objective was to establish the distance from the origin where the average cell size matches the chord length. This subdomain will be referred to as *inner far field* throughout this report.
To identify the ideal distance of the *inner far field*, four simulations were conducted with varying radii: 15, 30, 60, and 120 chord lengths from the origin. The same cell size was applied to the profile of each simulation, with a one-chord spacing enforced at each *inner far field* boundary. The criterion for selecting the distance was based on the evaluation of the relative error of C_L, C_D and C_M between consecutive simulations: the threshold considered satisfactory was set at 2% for C_D. This resulted in a distance of 60 chord lengths for the *inner far field*.
After defining the *inner far field*, the next step was to establish the far field distance. This involved running three additional simulations, each with varying far field distances and spacings while keeping the inner far field fixed. The first simulation used a two-chord spacing (at 120 chord lengths distance), the second a four-chord spacing (at 240 chord lengths distance), and the third an eight-chord spacing (at 480 chord lengths distance). A similar criterion was used, setting a relative error threshold of 0.5% for C_D. This resulted in a far field distance of 240 chord lengths, with a spacing of four chord lengths. The results obtained from this study are presented in Figure 2.

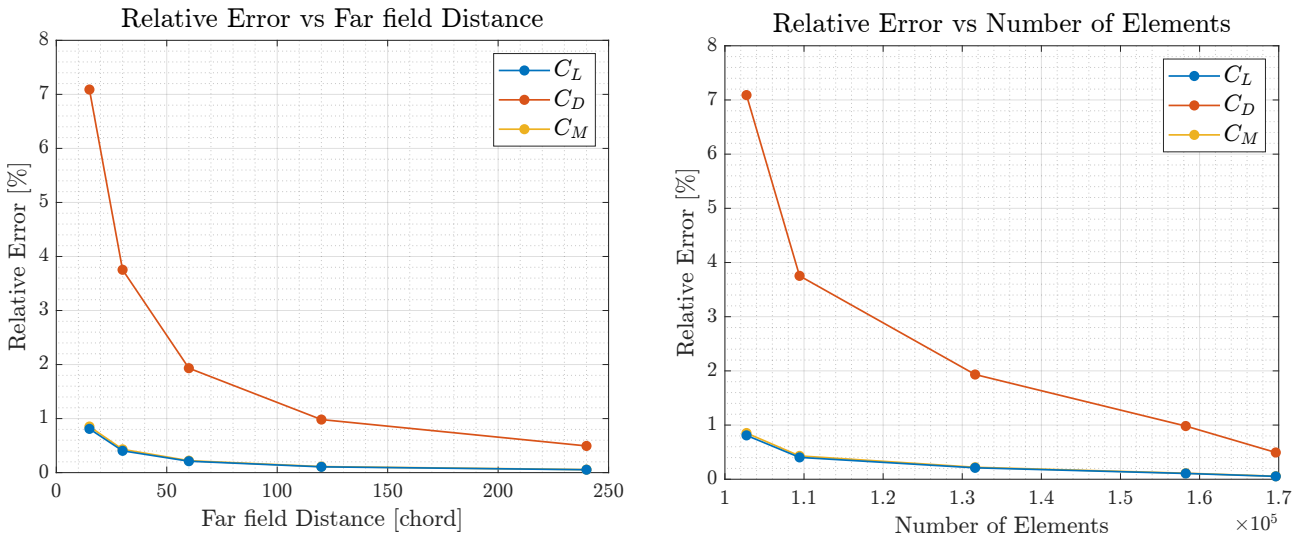


Figure 2: *inner* and *outer far field* independence studies. The first three points are related to the *inner far field*, whereas the other data are obtained by fixing the *inner far field* and adding the far field at 120 chords distance (with spacing of two chords lengths), at 240 chords distance (four chords lengths) and at 480 chords distance (eight chords lengths).

2.1.2 Element distribution on the profile

The second phase aims to define the surface meshing technique. The main aspects that have been taken into account are the necessity of describing the complexity of the iced geometries and of producing a refined mesh in the boundary layer to accurately resolve the strong gradients characteristic of this region, ensuring precise representation of physical phenomena.

The first consideration influences the dimension of the elements on the profile, with particular emphasis on the LE, where ice formation introduces complex geometrical features. The initial spacing was guided by insights from [3]. As shown in Figure 3, the mesh is progressively refined towards the leading and trailing edges, where flow gradients are most significant. In particular, there is strong curvature in the LE and a singularity in the TE, hence the necessity to refine. Specifically, a variation in spacing is applied from the leading edge to the center, and subsequently to the trailing edge. On the other hand, for iced profiles the spacing at the LE is maintained constant along all the region that presents a strong variation in the geometry of the ice structures.

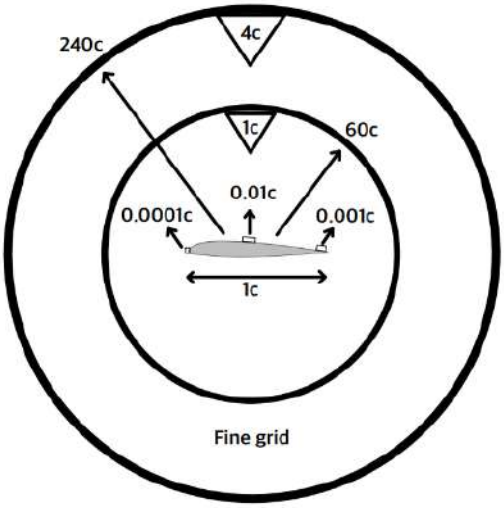


Figure 3: Mesh dimensions

To ensure a precise approximation of the viscous term, and consequently of the viscous drag, a y^+ value of 1 is initially adopted. This choice establishes a sufficiently dense grid within the boundary layer, allowing for an accurate resolution of flow gradients. However, reducing y^+ further improves accuracy, but it also raises the risk of discretization issues due to overly fine mesh resolution. By setting $y^+ = 1$ and determining the initial cell height within the boundary layer, the required mesh resolution is achieved. All other properties of the design of the boundary layer mesh are related to the mesher used. Both Gmsh and Pointwise have been used as meshers in this study: while Gmsh was initially used, Pointwise was introduced to address specific challenges that Gmsh could not overcome.

2.1.3 Boundary layer mesh generation challenges in Gmsh for iced airfoils

Issues related to the generation of the boundary layer meshes are due to the limitations in the extrusion algorithm of Gmsh. This algorithm leads to the extrusion of the boundary layer cells in direction normal to the line loop defined in the `.geo` file. For the iced airfoils studies, the complex geometry of the iced LE worsen these challenges.

The complex, non-uniform surface structure of the ice accretion, leads to overlapping cells during the extrusion process, causing the violation of Gmsh's meshing constraints. In order to overcome this problem, the airfoil profile was segmented into two regions: the first extends from the TE to the beginning of the iced-region, and the second covers the entire iced region. This segmentation allows for the definition of two independent thicknesses. Specifically, the structured mesh thickness in the frozen zone has been manually adjusted to avoid cell overlap errors caused by Gmsh's extrusion algorithm. However, this approach introduced a discontinuity at the interface, causing significant element stretching and high skewness in the unstructured triangular mesh, negatively affecting simulation accuracy and stability.

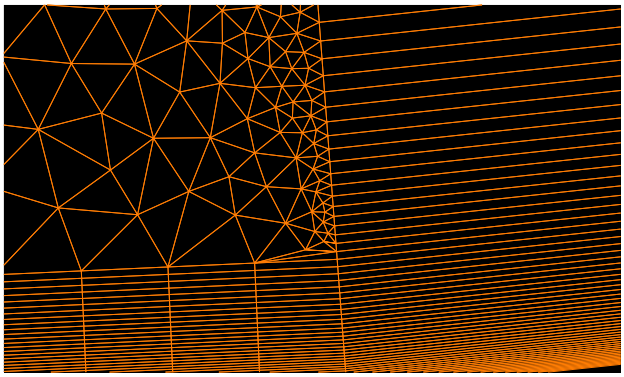


Figure 4: Effects induced by the discontinuity of the boundary layer: strong variations in terms of size of the cells and stretched triangular cells are highlighted.

2.1.4 Mesh generation using Pointwise

As a result of the limitations previously explained, Pointwise has been introduced as a mesh generation software. Initially a fully unstructured mesh has been generated using the Advancing Front algorithm.

A decay parameter of 0.8 was applied to enforce gradual refinement near the airfoil surface, ensuring higher resolution in critical regions such as the boundary layer and wake to accurately capture flow characteristics. The T-Rex (anisotropic extrusion) method was then employed to generate structured, BL-resolving cells. T-Rex method takes as inputs mainly two parameters. The first is the first-cell height to satisfy the non-dimensional wall distance ($y^+ < 1$), requirement for accurate BL modeling. The other is a specific growth rate to ensure smooth layer progression. The maximum number of layers was set to an arbitrarily large value, allowing the T-Rex algorithm to autonomously determine the termination of quadrilateral anisotropic layers, based on local geometric and spatial constraints. Beyond this point, the mesh transitioned to an isotropic triangular unstructured region for far-field discretization.

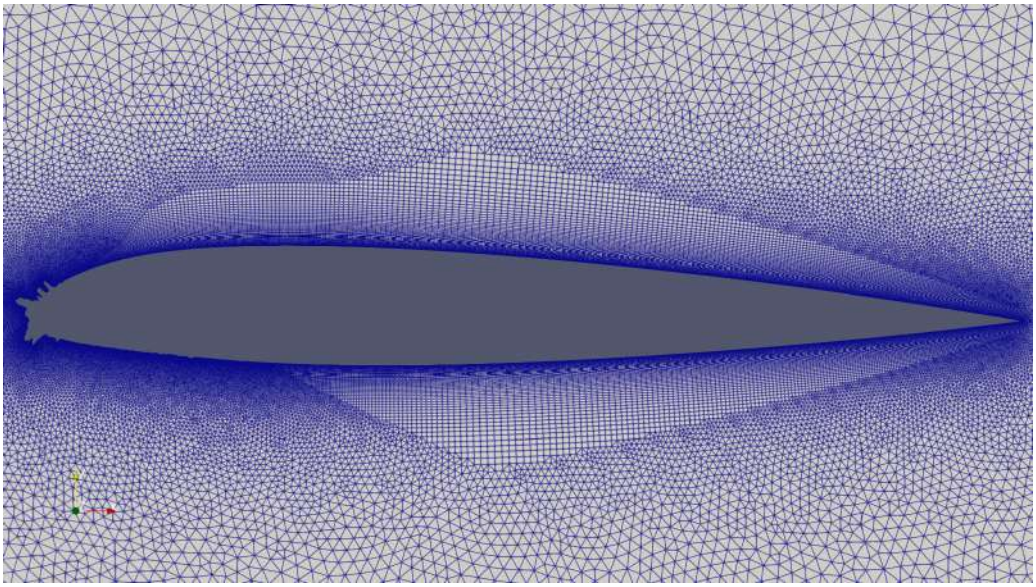


Figure 5: A view of the mesh near the profile (Case-251), generated using the T-Rex algorithm. The structured mesh region presents a smooth variation in thickness along the chord: higher-quality meshing is ensured by the absence of discontinuities or abrupt steps.

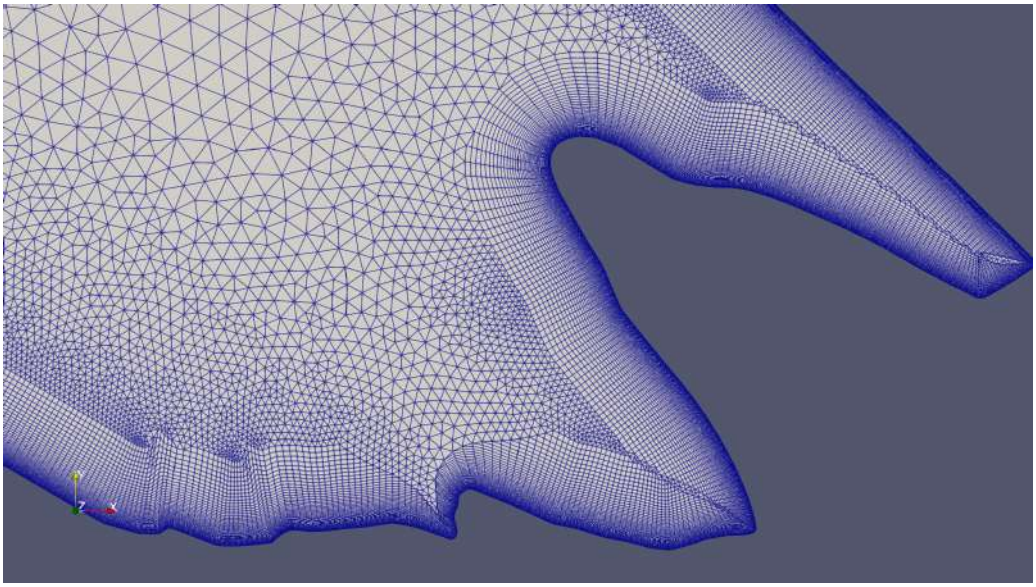


Figure 6: An up-close view of the mesh in a region with complex geometry.

2.1.5 Grid Convergence

After specifying the far field domain size and cell sizes (as described in Section 2.1.1), simulations were conducted using three progressively refined meshes: coarse, medium, and fine. The quantities of interest, C_L and C_D, were evaluated to assess mesh convergence. Following the guidelines outlined in [5], the mesh refinement ratio must be at least 1.1 to differentiate discretization error from other error sources. For most cells, a refinement ratio of $\frac{1}{1.4}$ was used, ensuring that cell area almost halves between fine and medium, and between medium and coarse meshes. A distinct method was applied to define growth rates within the boundary layer as follows:

- before the introduction of T-Rex, a fixed total height of the boundary layer (h_{tot}) was computed and kept across all meshes; similarly, the first cell height (h_1) was fixed to satisfy the y^+ condition;
- using T-Rex, despite its independent imposition of h_{tot}, the validity of the assumption $h_{tot}/h_1 \to \infty$ in the BL ensures the coherency of the equation;
- the last cell height in the structured layer was adjusted in order to have the same refinement ratio of the rest of the mesh. To fulfill this condition, here is the computation of a new growth rate (r_{new}) starting from a reference one (r_{ref}), where f is the chosen refinement ratio:

$$r_{new} = \frac{r_{ref}(h_{tot}/h_1 - 1)}{r_{ref}h_{tot}/h_1 - f[1 - (1 - r_{ref})h_{tot}/h_1]} \tag{1}$$

Choosing $r_{ref} = 1.1$ for the fine mesh, $r_{new} = 1.146$ and $r_{new} = 1.217$ result for medium and coarse meshes. The refinement ratios, calculated as the ratio of average cell sizes between consecutive meshes, are 1.2835 (fine to medium) and 1.2414 (medium to coarse), demonstrating independence. For results to fall within the asymptotic convergence range, two *Grid Convergence Index* (GCI) values were computed across the three grids. The fine-grid GCI values serve as error estimates for the plotted results, in accordance with the method outlined in [9]. The results obtained for the clean airfoil, including Richardson extrapolation, are presented in Figure 7. The corresponding GCI values are expressed as percentage errors, with $GCI_{fine_{C_L}} = 0.00046\%$ for C_L and $GCI_{fine_{C_D}} = 0.0198\%$ for C_D. These low GCI values indicate that the numerical results are highly reliable, with minimal discretization error, confirming the adequacy of the mesh refinement levels for the analysis of the clean airfoil.

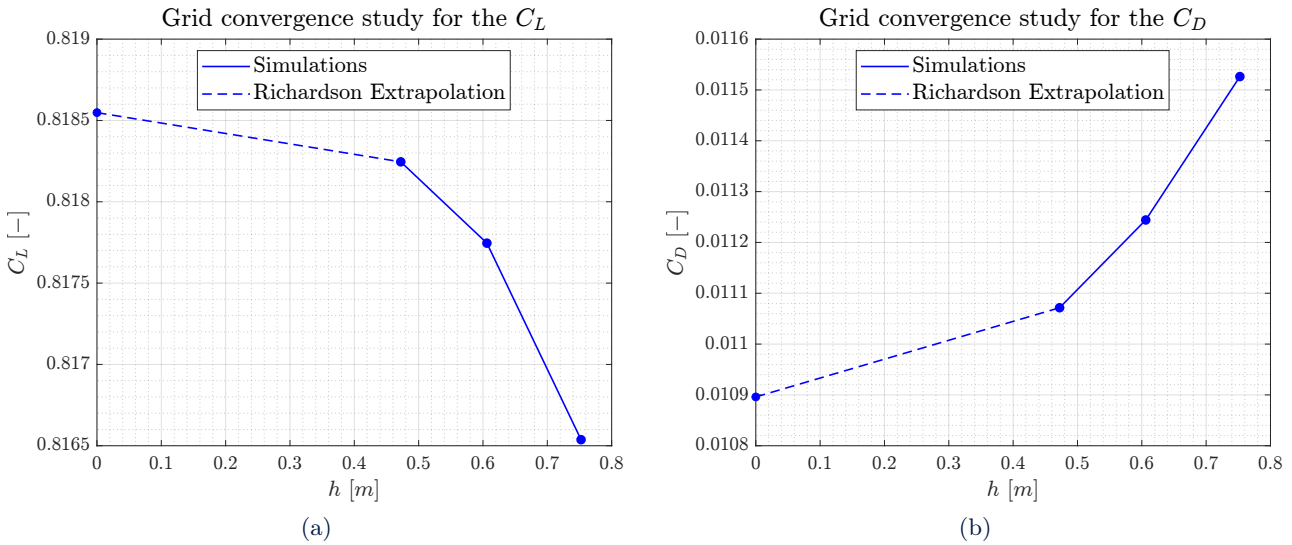


Figure 7: Results of the grid convergence study for the clean NACA 23012 airfoil, showing the comparison of aerodynamic coefficients (C_L and C_D) across progressively refined meshes.

This calculation could not be verified for iced profiles due to the complexity of such geometries. Even minor variations in mesh characteristics lead to oscillations in results between consecutive meshes, preventing the establishment of asymptotic convergence, as presented in [8]. Despite these challenges, a convergence study was conducted for all cases. A further refined mesh was chosen for Case-251, given its complexity, to reduce these limitations and to improve the accuracy of the results.

	Clean	Case-241	Case-242	Case-251
N° Elements	169,629	519,334	308,537	856,724

Table 1: Number of elements used among different simulations. Despite the simpler geometry, it can be seen that the number of elements is larger in Case-241 than in Case-242. This is attributable to the fact that the ice-covered zone begins further aft in Case-241 than in -242 (see Figure 1). This, combined with the method used for the discretization, leads to an increase in the number of cells.

2.2. Numerical methods setup

2.2.1 Governing equations

To identify a suitable analytical framework for the governing equations, the physical problem was rigorously analyzed with a focus on the dominant flow phenomena.
The original freestream conditions of Case-241, from [1], were selected as the baseline for comparative analyses of the results. Given the experimental Mach number, it was considered whether to use the incompressible or compressible governing equations. Since $M > 0.3$ in all cases, compressibility effects were deemed significant, leading to the adoption of the compressible flow equations. Additionally, the effects on the flow caused by the geometry of the ice layers were evaluated. The intricate structures visible near the LE, including complex surface roughness, have a substantial impact on the flow, particularly through phenomena like boundary layer separation, recirculation bubbles, turbulent transition, and reattachment. To model the flow field, 2D compressible RANS equations were selected. Section 2.2.2 describes the turbulence modeling approach.

2.2.2 Turbulence models

Several models provide closure for the Reynolds-stress tensor in RANS equations, differing primarily in how eddy viscosity is computed. This study focuses on the Spalart−Allmaras (SA) and $k-\omega-SST$ models, as they are the only options available in the SU2 solver.

- Spalart−Allmaras (SA): this model is intended for aerodynamic flows, such as transonic flow over airfoils, possibly featuring boundary layer separation. However, it is overly dissipative in regions with high streamline curvature. Given the expected prevalence of recirculation bubbles in the studied cases, the rotational correction was enabled.
- $k-\omega-SST$: this model combines the strengths of the $k-\epsilon$ and $k-\omega$ models via a blending function incorporated into the ω equation. Near walls, the model behaves like a standard $k-\omega$ formulation, while it transitions to a $k-\epsilon$ behavior in the free stream.

Additionally, the sensitivity of the solution to transition modeling was investigated by coupling the SA model with one of the two transition models available in SU2 at a time: the Boundary Layer Correlation-based Model (BCM) and the Langtry-Menter (LM) model.

2.2.3 Boundary and freestream condition

This section offers an overview of the main settings of the configuration file, such as free-stream conditions, boundary conditions, and convergence criteria for the simulations.
The freestream parameters were specified as follows:

- Freestream Mach number $M = 0.325$;
- Freestream pressure $P_\infty = 92,528\ Pa$;
- Freestream temperature $T_\infty = 250.15\ K$;
- Reynolds number $Re = 3.8 \times 10^6$ (using the airfoil chord as the reference length);
- Angle of Attack: varies depending on the specific study point.

The Sutherland viscosity model, with standard reference parameters for air, was used to capture temperature-dependent variations in dynamic viscosity. A constant Prandtl conductivity model was used, with conventional air values. The circular mesh eliminates the need to divide the far field into inlet and outlet, permitting freestream conditions along the whole boundary. Wall boundary conditions

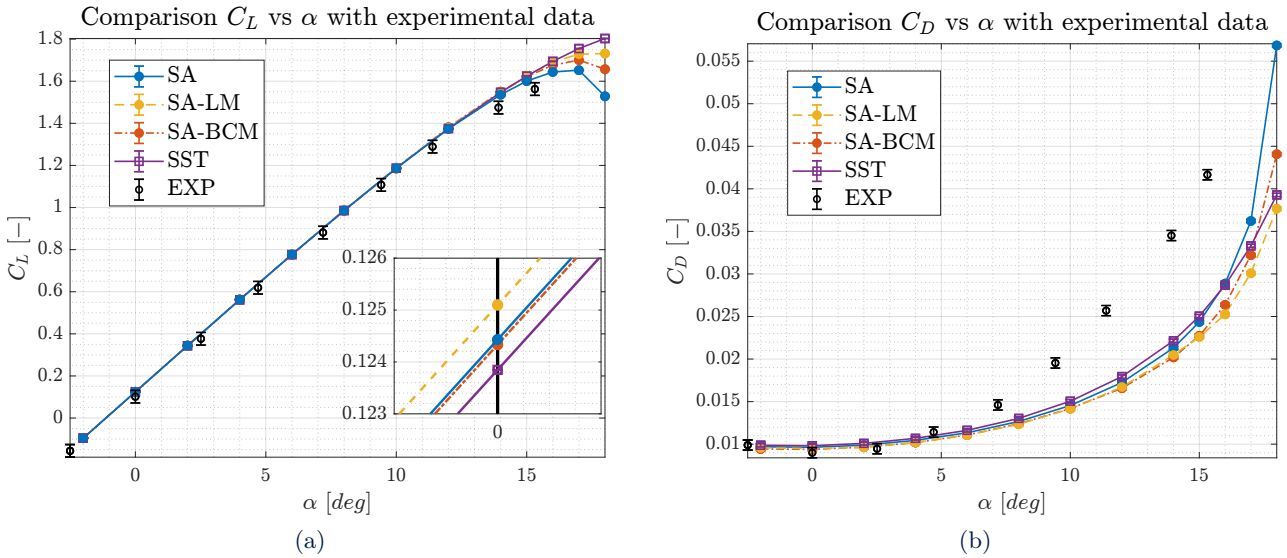
on the airfoil surface required marker heat flux for the RANS solver. An adaptive *CFL* number was employed to both accelerate simulations and mitigate instabilities. Convergence was assessed using the root-mean-square density residuals (rms_ρ) and the Cauchy-averaged lift ($Cauchy_{C_L}$) and drag ($Cauchy_{C_D}$) coefficients to ensure robust aerodynamic evaluation. The convergence thresholds were set as follows: $rms_\rho = -10.5$, $Cauchy_{C_L} = -6$, $Cauchy_{C_D} = -6$. Additional settings included the use of the *Weighted Least Squares* method for spatial gradient computation, the *FGMRES* linear solver (with 25 inner iterations and *ILU* preconditioning for enhanced stability), the *JST* convective numerical scheme (preferred over the Roe scheme due to convergence challenges encountered with iced airfoils), and the *Implicit Euler* scheme for advancing computations in a pseudo-time framework.

3. Results

3.1. Clean airfoil and validation

3.1.1 Experimental validation

To check the reliability of the computational results, the CFD simulations were first validated by reproducing selected wind tunnel experiments conducted by Harris (1939) [7]. According to the original report, these experiments were performed under standard conditions at a nominal Reynolds number of $Re = 2.19 \times 10^6$. However, due to the high turbulence levels in the wind tunnel, the effective Reynolds number was approximately $Re = 3.5 \times 10^6$. Given the considerable uncertainties associated with the experimental conditions and the measurement of aerodynamic coefficients – reported as 1.92% for $C_{L_{\max}}$ and 3.33% for $C_{D_{\min}}$ – only the lift coefficient (C_L) was used as validation metric. However, the predicted curves for both lift (C_L) and drag (C_D) coefficients are presented in Figure 8 for completeness. Figure 8a provides a detailed view around $\alpha = 0°$. The results show that all CFD simulations fall within the experimental error margins across the linear portion of the lift curve, demonstrating strong agreement with the experimental data and validating the computational approach.



(a) (b)

Figure 8: Comparison of the CFD results with Harris (1939) [7] experimental wind tunnel data. The lift coefficient and drag coefficient are shown, highlighting the agreement of C_L between simulated and experimental results within the error margins.

3.1.2 Clean profile results

This section presents the simulation results obtained using the boundary and freestream conditions described in Section 2.2.3. Figures 9a and 9b show that the results match closely throughout the linear portion of the curves, with an almost perfect overlap up to $\alpha = 10°$. This behavior is consistent with

expectations, as the flow remains fully attached in this range, showing no signs of recirculation zones. At higher angles of attack, differences start to appear due to the onset of mean flow separation at the trailing edge of the profile. The simulations using the Spalart−Allmaras (SA) turbulence model consistently predict a maximum C_L at $\alpha = 15°$, followed by a sharp drop in lift, culminating in deep stall conditions at approximately $\alpha = 20°$. In contrast, the $k-\omega-SST$ model shows a less drastic drop in performance after stall. This divergence arises from differences in how the two models handle near-wall flow physics. As illustrated in Figure 10, at $\alpha = 18°$, the SA model predicts a much larger separation bubble, indicating a deeper stall. This results in lower C_L and higher C_D.

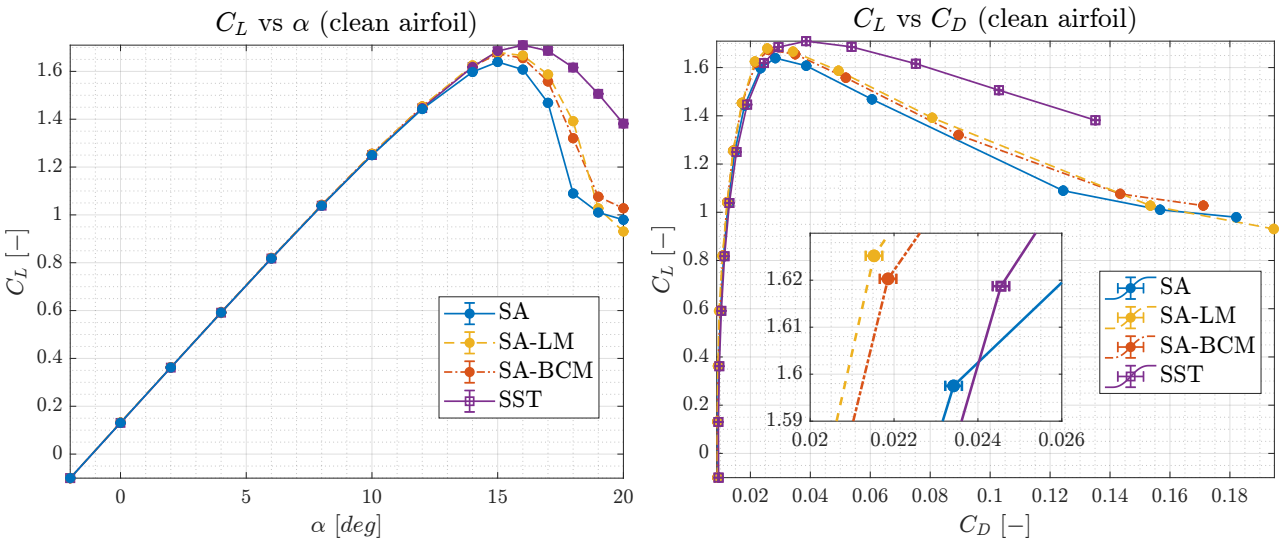
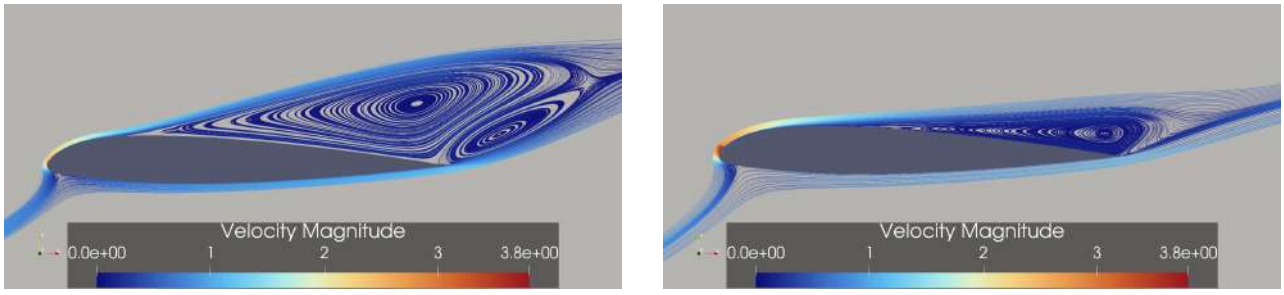


Figure 9: Lift coefficient curve and drag polar of the clean NACA 23012 airfoil, comparing results from SA turbulence model, SA combined with the transition models, and the $k-\omega-SST$ turbulence model.



(a) Recirculation bubble obtained using SA

(b) Recirculation bubble obtained using $k-\omega-SST$

Figure 10: Comparison of the recirculation bubble regions obtained from SA and $k-\omega-SST$ turbulence models with the clean NACA 23012 airfoil at an angle of attack of $\alpha = 18°$. The velocity magnitude is normalized such that a value of 1 corresponds to the velocity set in the far field.

3.2. Case-241

Case-241 focuses on rime ice, which mostly increases the effective chord length because of its overall smooth and regular surface on a macroscopic scale. Despite this, the surface has pronounced waviness, with crests and valleys reaching up to one-third of the clean airfoil's chord length. These undulations induce a series of near-wall recirculation zones. However, the overall impact on aerodynamic performance is less severe compared to other cases. Figure 11 presents the simulation results, which closely

resemble trends observed for the clean airfoil configuration. Simulations based on the SA turbulence model predict a lower maximum lift coefficient $C_{L_{\max}}$, an earlier stall angle, and a steeper post-stall drop in lift compared to those based on the $k-\omega-SST$ model. Interestingly, in contrast to the clean airfoil results – where SA and SA coupled with transition models ($SA-BCM$ and $SA-LM$) showed qualitatively similar but quantitatively distinct behavior – Case-241 demonstrates nearly identical predictions across SA, $SA-BCM$, and $SA-LM$. This observation is highlighted in the magnified insets of Figure 11. This is due to the flow's forced laminar-to-turbulent transition caused by numerous separation and reattachment zones near the leading edge. Moving past the forward area near the stagnation point, the damping effect of the transition model is disabled, allowing the rest of the flow field to be fully turbulent [6]. As a result, the solutions obtained with SA, $SA-BCM$, and $SA-LM$ models converge to nearly identical values.

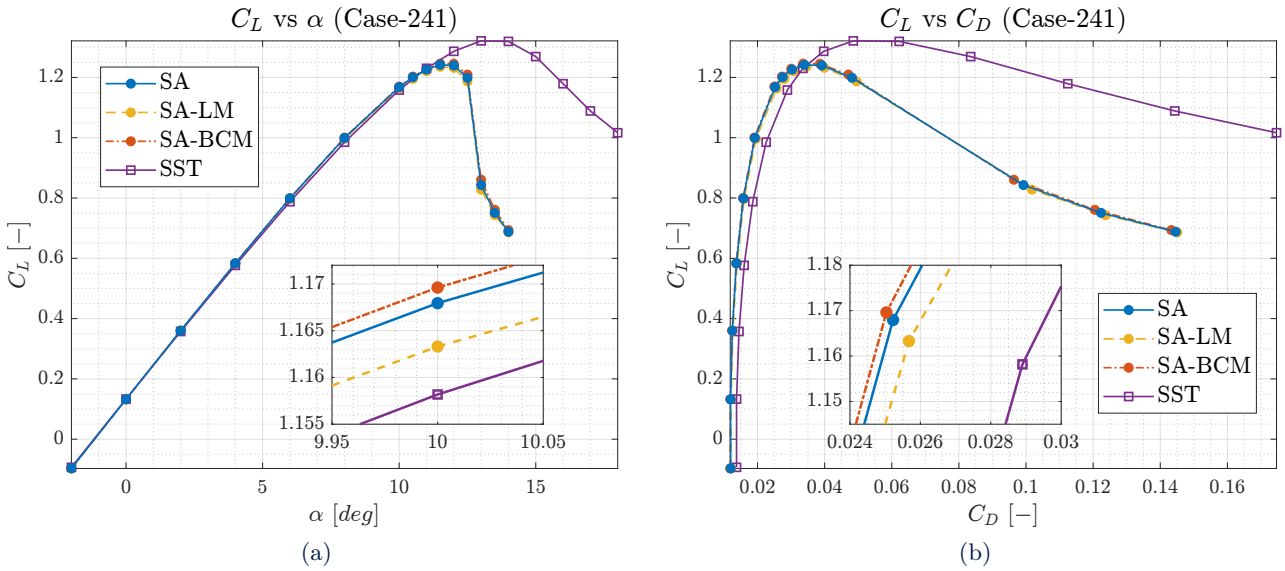


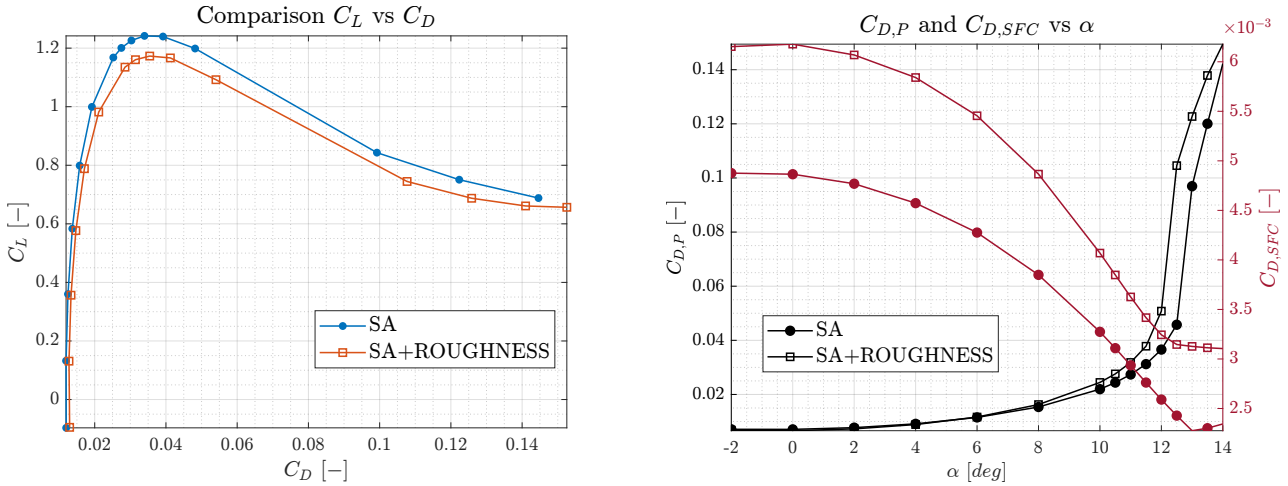
Figure 11: Lift coefficient curve and drag polar of Case-241, comparing results from SA turbulence model, SA combined with the transition models, and the $k-\omega-SST$ turbulence model. A magnified inset highlights the minimal difference between the SA model variants

3.2.1 Case-241 with small-scale ice roughness

The analysis for Case-241 simplifies real-world conditions by focusing mainly on geometric changes to the airfoil and initially ignoring the effects of small-scale surface roughness. In reality, the performance degradation of an iced airfoil results from a combination of shape alterations and increased surface roughness, the latter of which has now been incorporated into the analysis. To account for surface roughness, an equivalent sand-grain roughness height (k_s) was applied exclusively to the iced region, while the rest of the airfoil surface was assumed to remain perfectly smooth. Following the methodology outlined in [4], the roughness value for the `.config` file was determined using Equation (2), where f_0, the freezing fraction at the stagnation point, equals 1 for rime ice.

$$\frac{k_s}{c} = \frac{1}{1000}\frac{1}{2}\sqrt{0.15 + \frac{0.3}{f_0}} \tag{2}$$

Figure 12a confirms that iced profiles with roughness experience performance degradation. A more detailed analysis is provided by decomposing the total drag coefficient (C_D) into its pressure-induced ($C_{D,P}$) and friction-induced ($C_{D,SFC}$) components. Both drag components are higher when roughness is applied. In particular, $C_{D,SFC}$ shows a nearly constant 30% increase, which is indicative of an increased turbulent activity in the near-wall region, which leads to higher skin friction drag, as shown in Figure 12b.



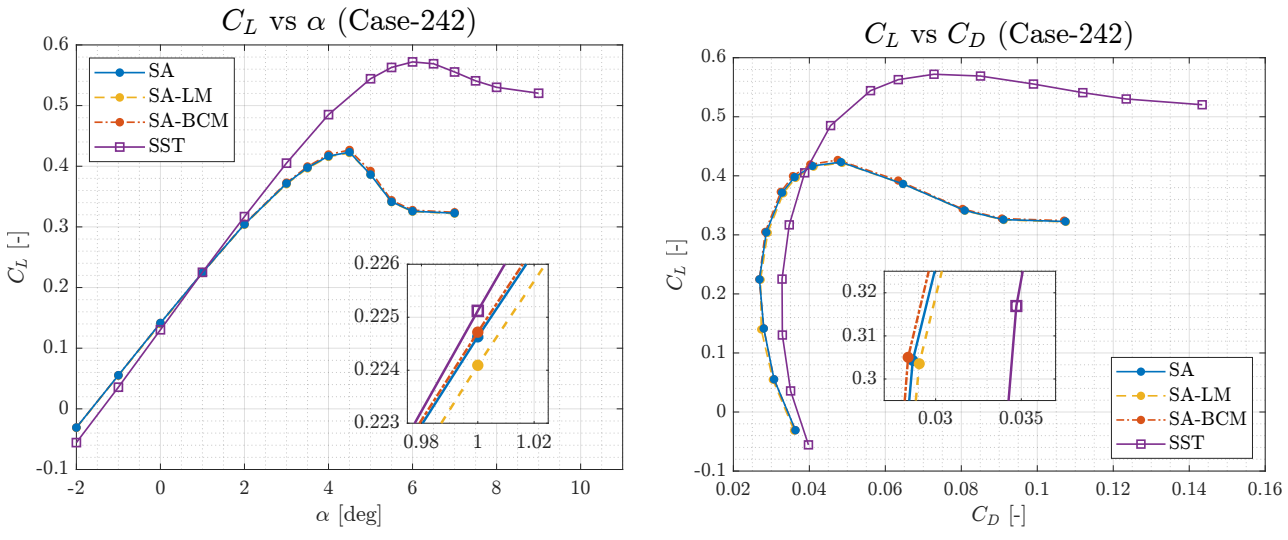
(a) Drag polar

(b) De-coupled drag coefficient curves

Figure 12: Drag polar (a) and the decomposition of drag components into pressure-related ($C_{D,P}$) and friction-related ($C_{D,SFC}$) (b) of Case-241 with and without surface roughness applied, using the SA turbulence model.

3.3. Case-242

Case-242, with glaze ice, is characterized by two large ice structures at the leading edge, resembling two horns. Alongside these two structures, other small, pointed ice shapes develop at the front of the profile. This complex geometry makes this case the most harmful in terms of airfoil performance compared to the other iced cases. Stall angle is approximately 4.5° according to SA model, while with $k-\omega-SST$ predicts a slightly higher value around 6°. Examining the C_L - α curve, the $k-\omega-SST$ model shows a steeper slope in the linear region compared to SA. At equivalent angles of attack, C_D value is also higher according to $k-\omega-SST$. Like for Case-241, very small variations can be observed among the various SA options tested. According to SA model, at $\alpha = 4°$ recirculation bubbles form only behind ice horns that then reattach downstream. At $\alpha = 5°$, just beyond stall, a recirculation bubble develops on top of the airfoil near the TE, merging with the bubble forming behind the ice horn at $\alpha = 7°$.



(a)

(b)

Figure 13: Lift coefficient curve (a) and drag polar (b) of Case-242. Two turbulence models are presented both with two transition models coupled to SA. The magnified inset emphasizes the minimal distinction between SA coupled with transition models and SA.

3.4. Case-251

Case-251 exhibits distinctive features of glaze ice, namely prominent horns, and additional major structures downstream from the stagnation point, together with bumps that extend up to about one-third of the chord length along the pressure side. This abundance of protrusions causes several cavity flows to appear at the LE and intricate vortical motion ensues (Figure 14). A main recirculation bubble near the LE on the pressure side is the most noticeable flow feature at low α, while a comparable separation zone on the suction side becomes apparent as α increases.

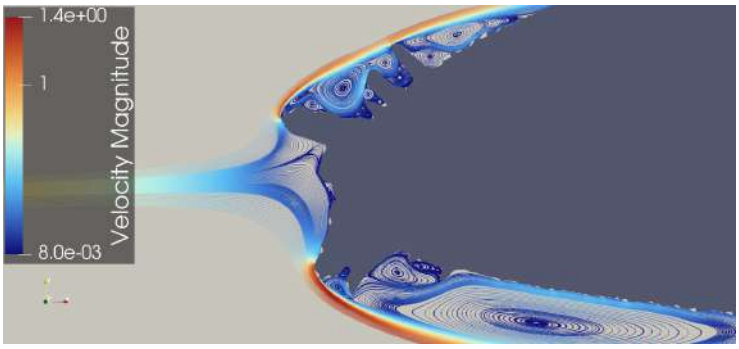


Figure 14: Streamlines at the leading edge of Case-251 (standard SA, $\alpha = 0°$). Velocity magnitude is normalized such that a value of 1 corresponds to the freestream velocity.

TE separation is first observed at $\alpha = 6°$ and the lift coefficient (Figure 15a) indicates that stall occurs when $\alpha = 7°$ according to SA model, and when $\alpha = 8°$ according to $k - \omega - SST$. This trend aligns with observations from previous cases. No discernible discrepancies between SA and its variants have been noted with respect to convergence behavior and simulation results, besides the inability of $SA - BCM$ to converge at negative α. All models led to considerably slow convergence when $\alpha = 8°$: although unsteady RANS should be considered for meaningful results concerning post-stall behavior, this is shown to be the point at which the LE separation zone and the TE one make contact and merge.

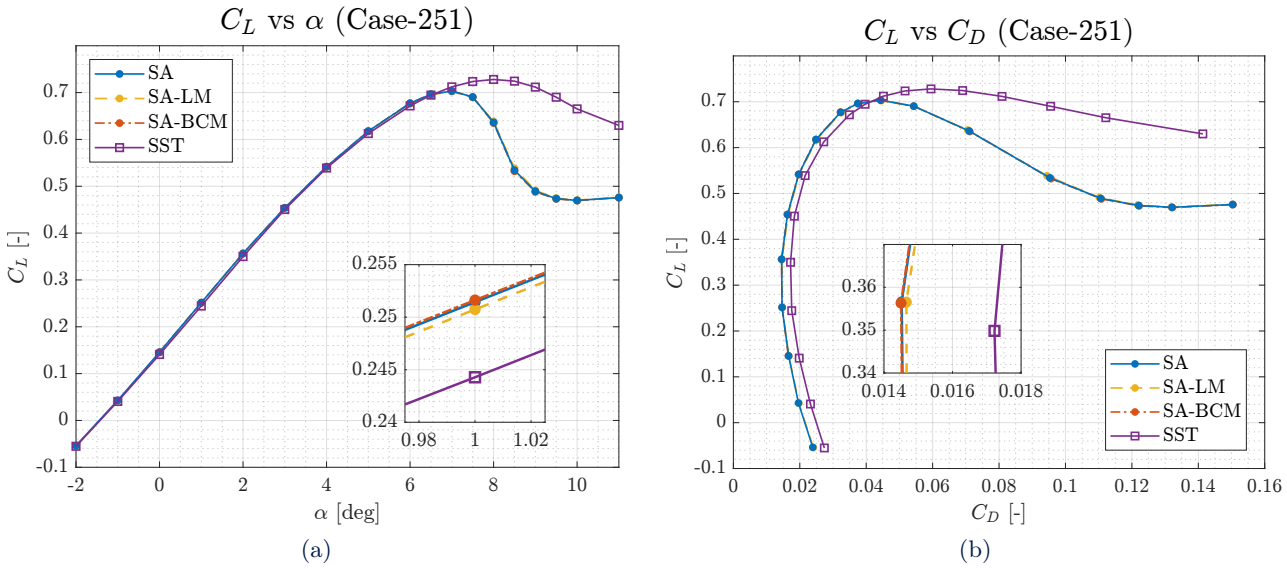


Figure 15: Lift coefficient curve and drag polar of Case-251, including all different turbulence models that were tested. A magnified tile highlights the negligible difference between SA variants.

3.5. Performance decrease due to ice accretion

All of the different geometries are discussed in the following, where reference will be made to the results of simulations run with SA turbulence model (not coupled with a transition model).

In general, rime ice (Case-241) proves to be less impactful than glaze ice, causing α_{stall} to drop from 15° to just shy of 12°, while the increase in C_D is moderate as long as the C_L behavior is linear. Conversely, glaze ice results in substantial performance deterioration, with a steep reduction in α_{stall}, as seen in Case-242 ($\alpha_{stall} < 5°$) and Case-251 ($\alpha_{stall} = 7°$). In addition, glaze ice is responsible for substantial

reductions in $\frac{\partial C_L}{\partial \alpha}$ derivative, as highlighted in Figure 16a. An increase in zero-lift coefficient C_{L0} is also observed in iced airfoils but it is an effect of minor importance. Perhaps unexpectedly, Figure 16 shows how C_L and C_D are worse in Case-242, despite the more complex ice shape of Case-251: the rationale behind this phenomenon might be related to secondary horns of the latter, located downstream of the main ones, which hinder (in part) surging of LE separation regions; this contribution and the influence of comparatively larger ice structures of Case-242 add up, leading to most noticeable departures from the clean-airfoil lift curve and drag polar.

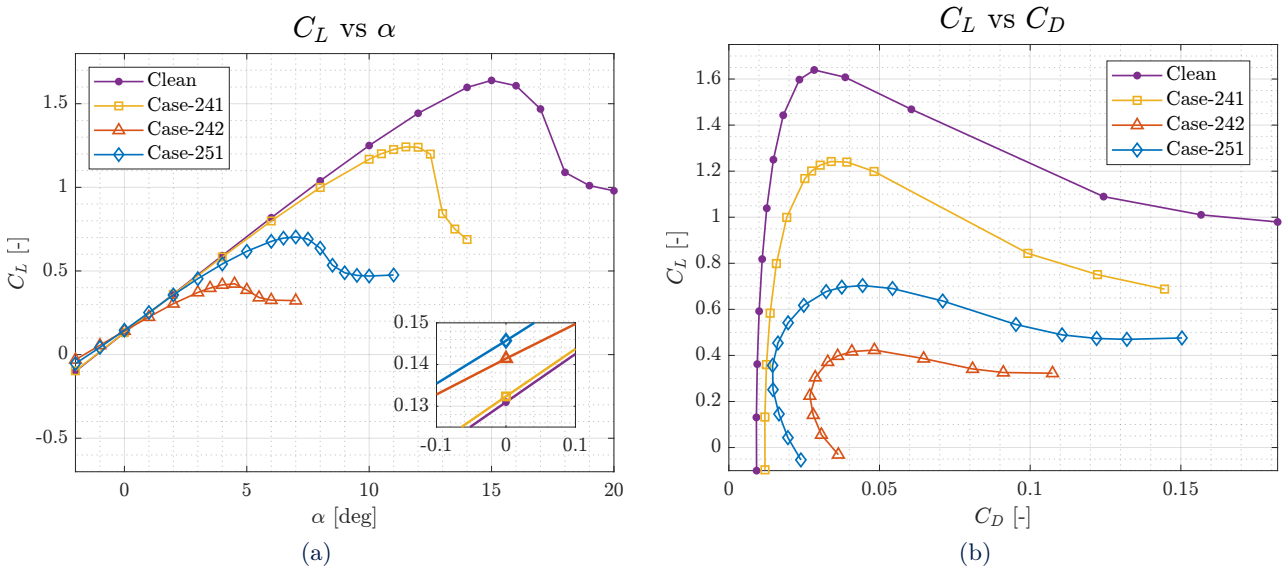


Figure 16: Comparison among lift curves (a) and drag polars (b) of all investigated airfoils. Results shown were obtained with standard SA model (rotational correction).

Regarding the pitching moment (C_M), there are significant changes between the different ice geometries (see Figure 17). Following `.config` files provided to SU2, C_M was computed at the quarter of chord. Trends of C_M curves seem to be qualitatively consistent with the findings reported in [2].
The quarter-chord point is approximately the aerodynamic center of the clean airfoil. However, for iced geometries, this shifts, as shown in Table 2. To shift the pole about which C_M is computed from $c(\frac{1}{4}, 0, 0)$ to $c(\xi, 0, 0)$, the following formula has been used (note that the sign of C_M has been changed with respect to SU2 output, to get positive pitching moment around $-z$, that is if the airfoil tends to pitch up):

$$C_{M,\xi} = C_{M,1/4} + (\xi - \frac{1}{4})\cos(\alpha)C_L + (\xi - \frac{1}{4})\sin(\alpha)C_D \tag{3}$$

Then, the aerodynamic center (AC) has been found as the point for which the slope of the regression line of the C_M - α curve is equal to 0. Not the whole C_M - α curve has been considered for the linear fit, but points from $\alpha = -2°$ to $\alpha = 8°$ for clean airfoil, $\alpha = 6°$ for Case-241, $\alpha = 4°$ for Case-242, $\alpha = 5°$ for Case-251 (covering approximately the linear region). Here is a summary with the position of the aerodynamic center in terms of chord fraction and $C_{M,AC}$ for each geometry:

		Clean	Case-241	Case-242	Case-251
ξ_{AC}	[-]	0.242	0.220	0.101	0.181
$C_{M,AC}$	[-]	-0.0106	-0.0133	-0.0300	-0.0255

Table 2: Estimated position of the AC and pitching moment coefficient of each geometry

For the clean airfoil, AC is actually close to 25% of the chord, but as the ice shape becomes more invasive – from Case-241 to Case-251 and Case-242 – the aerodynamic center moves closer to the leading edge.

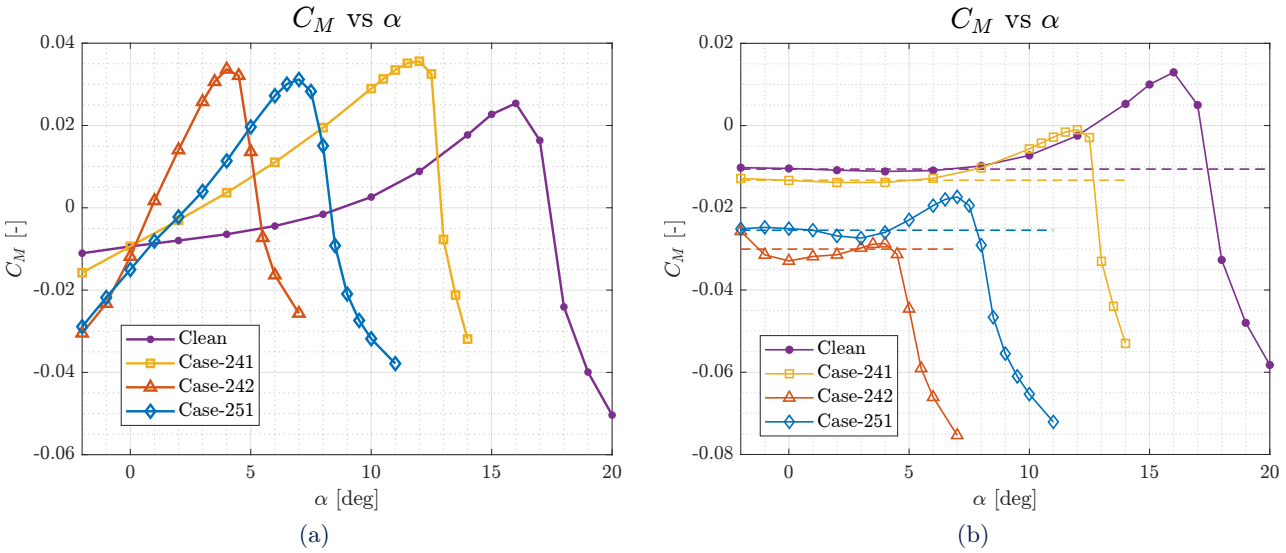


Figure 17: Comparison among pitching moment curves, with respect to $\frac{c}{4}$ (a) and AC (b), of all investigated airfoils. Results shown were obtained with standard SA model (rotational correction).

3.6. Parallel study of a mesh created by a MATLAB® script + Gmsh

As mentioned above (Section 2.1.3), mesh generation with Gmsh for iced geometries brings about several problems, which is why Pointwise was preferred. But in parallel with learning how to create a mesh with Pointwise, another approach has been tried with Gmsh.

The primary issue with Gmsh arises when creating boundary layer structured cells: where there are sharp corners, with the angle between two consecutive edges on the airfoil surface that is much smaller than 180° (the one measured externally to the airfoil), extrusion of layers may raise errors, like in the bottom right of Figure 18a (close-up of an ice irregularity). A potential solution would be to manually reduce the total height of the boundary layer cells near these points, though this would be a time-consuming process for each corner where such errors occur. Given that there are numerous points like this throughout the iced region of Case-251, and considering the need to test grid convergence, this manual process would have to be repeatedly carried out.

Additionally, Gmsh built-in function for the boundary layer structured meshes presents issues in corners like the one on the left of Figure 18a, where instead the angle between two consecutive edges (considering the one external to the airfoil) is much greater than 180°. In these cases, cell width becomes quite large, making it difficult to achieve a uniform aspect ratio between cells. While Gmsh does offer a fan option for the BL – such as at the trailing edge of a standard airfoil – this process requires manual intervention for each point where a fan needs to be added.

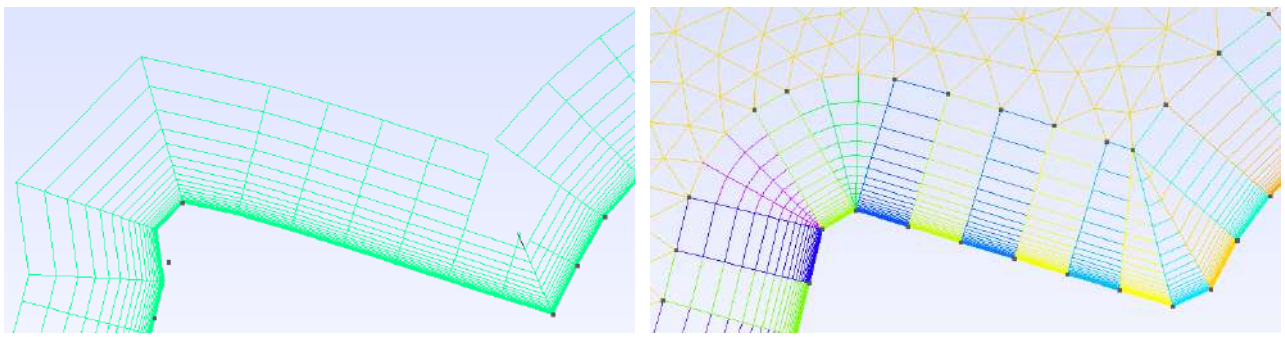


Figure 18: Comparison of boundary layer creation with Gmsh built-in function (a) and with MATLAB® + Gmsh (b)

Because of the reasons explained above, it was decided not to use the Gmsh built-in function to create boundary layer structured cells. In the following approach, a MATLAB® script was developed to automate the generation of the required `.geo` file, for Gmsh. Below is a summary of the steps involved in creating the `.geo` file:

- points on the airfoil surface are created interpolating the original database with the desired spacing, which varies along the airfoil for improved mesh control. An offset of these points is computed to define the external boundary of the structured mesh;
- lines normal to the airfoil are created to connect each interpolated point with its corresponding offset point. These lines are defined as transfinite lines with the desired progression rate and the number of points set to obtain the desired height of the first cell (to fulfill the y^+ condition). Lines parallel to the airfoil are defined to connect points on the airfoil surface and, separately, points on the external boundary of the structured mesh. These lines are defined as transfinite lines with only two points (to avoid that intermediate points are automatically created by Gmsh);
- in general, a line loop is created connecting four adjacent lines that form a quadrangle, that will then serve to delimit the plane surfaces, then set as transfinite surfaces. Each uniform-color region in Figure 18b represents a different surface;
- in the case where the offset points tend to converge like the one on the right of Figure 18b, a triangular line loop is created instead of a quadrilateral one to prevent Gmsh errors, like mentioned before. In the opposite case where consecutive offset points spread apart, like on the left of Figure 18b, fans are added;
- the unstructured external mesh up to the far field is defined using the same commands as in the Gmsh-only approach.

The difference in the resulting mesh can be appreciated by comparing Figure 18a and Figure 18b: the latter is successfully built by Gmsh without errors, it is more regular and it simplifies the creation of multiple meshes with different mesh sizes.

Here is a comparison of the drag polar of Case-251 (which features the most complex and irregular ice shape) between simulations run with the mesh made with Pointwise and the mesh made by Gmsh from the `.geo` file created by MATLAB®. The same `.config` file has been used, with SA as turbulence model.

While C_D values are nearly identical for both meshes at all angles of attack between $-2°$ and $7.5°$ (post-stall), C_L exhibits slight differences closer to stall. The overall trends of the two curves is exactly the same, but this shows how difficult it is to achieve mesh independence in the case of a complex iced geometry like Case-251: even a very small difference between the two meshes in a few points on the airfoil can make a difference on the results, even if the size of the cells between the two meshes is similar.

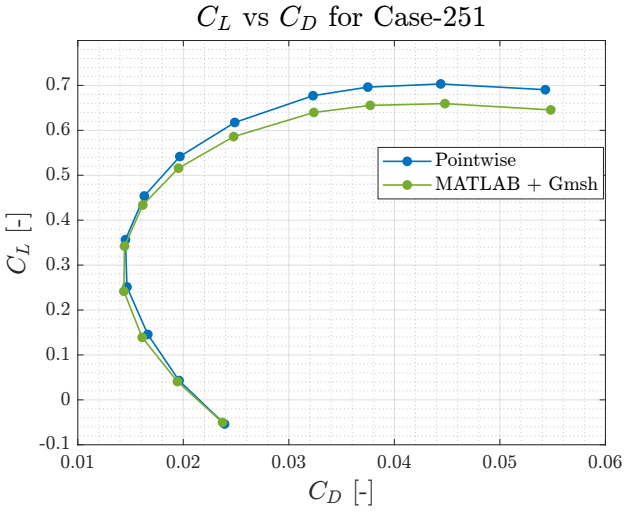


Figure 19: Drag polar comparison between mesh created by Pointwise and MATLAB® + Gmsh.

4. Conclusions

The primary concern caused by the presence of an ice layer over a lifting surface is the dramatic deterioration of its performance due to geometry changes affecting the LE. The geometry itself poses the first challenge of icing simulations when it comes to grid convergence, which cannot be easily achieved as clarified in Section 2.1.5. Regarding the meshing software, the use of Pointwise guarantees better mesh independence in general, but in case of limited resources, the usage of an open source software like Gmsh can also give good results if properly exploited.
Flow around ice shapes may feature several separation bubbles, something which causes profoundly different behaviors when distinct turbulence models are compared, more so than in the case of a clean airfoil. In addition, SA transition models are shown to entail no significant difference with respect to standard SA because of the flow's forced laminar-to-turbulent transition caused by numerous separation and reattachment zones near the leading edge.
The study confirmed not only that a significant increase in drag and a α_{stall} reduction should be expected, owing to a less streamlined shape, but also that a further major effect is linked to how the pitching moment of the airfoil is altered (Section 3.5); in fact, C_M is affected by ice accretion in such a way that $\frac{\partial C_M}{\partial \alpha}$, a close relative of the longitudinal stability derivative (to be instead computed with respect to the center of mass), gets larger at smaller values of α up to stall: this might be a factor responsible for critically affecting aircraft handling in icing conditions. On a final note, both Case-242 and Case-251 confirm that glaze (horn) ice accretion significantly compromises aeronautical applications.

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